Modeling of Isotope Distributions and Intracellular Fluxes in Metabolic Networks Using Atom Mapping Matrices

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The modeling of isotope distributions can be used to evaluate intracellular fluxes and to investigate cellular metabolism. A method of modeling isotope distributions in biochemical networks that addresses some of the shortcomings of conventional methods is presented. Matrix equations representing steady state isotope balances are formulated for each metabolite and solved iteratively via computer. The key feature of this method is the use of atom mapping matrices, which decouple the generation of the steady state equations from the details of the transfer of carbon atoms from reactants to products. The use of atom mapping matrices results in a clear, intuitive description of metabolic networks that is easy to develop, check, and modify. A network representing energy metabolism in a hybridoma cell line was developed and presented as an example of the method. A program that uses the atom mapping matrix method to calculate the isotope distribution in the network as a function of the intracellular fluxes was written. Calculations showed that a single nuclear magnetic resonance (NMR) experiment with ^{13}C-labeled glucose should be able to uniquely determine two important TCA cycle flux ratios. These results demonstrate how *in vitro* NMR can be used to study cellular metabolism or to validate the flux estimates obtained from other methods.

1. Introduction

The action and regulation of metabolic networks are complex because they involve a large number of enzymes and a wide variety of control mechanisms. Quantitative analysis of intracellular fluxes is an important tool for investigating cellular metabolism. Measurement or estimation of fluxes through a biochemical network, combined with measurements of intracellular concentrations or enzyme activities, can help to elucidate *in vivo* regulatory mechanisms and to evaluate environmental effects on cultured cells or tissue. Furthermore, flux analysis provides a way to control and optimize cell culture processes by characterizing the physiological state of the cell.

Intracellular fluxes can be determined by several methods. For example, fluxes through some pathways can be measured directly via *in vivo* nuclear magnetic resonance spectroscopy (NMR) using magnetization transfer or isotope transfer techniques (*1, 2*). However, direct flux measurement with NMR is of limited use because it is inherently insensitive and because it requires that the spin–lattice relaxation times of the reactants and products be of the same order as the reaction rate. Indirect methods utilizing isotopic tracers can be used to overcome the limitations of direct NMR techniques. In isotopic tracer methods, cells are provided with a substrate that has been labeled with a detectable isotope at a specific atom. A material balance is performed on the labeled isotope, and the measured isotope distribution among, and within, different metabolites is used with knowledge of the biochemical reactions to estimate the fluxes of interest. An alternative method measures the rates of change of extracellular metabolite concentrations and uses a total mass or carbon balance to calculate intracellular fluxes (*3*). Confidence in the resulting flux estimates can be increased by independent validation with isotopic tracer methods.

Flux estimates from isotopic tracer methods usually use substrate enriched with ^{14}C, which is radioactive, or ^{13}C, which is stable and detectable with NMR. Equations are formulated describing the steady state specific activity of each carbon atom of the metabolites. For radioactive isotopes, specific activity is usually expressed as radioactivity per mole or per gram. For stable isotopes, specific activity is usually expressed as the fractional enrichment of a specific atom within a molecule. The steady state equations relate the specific activities of the metabolite carbon atoms to the specific activities of the substrate carbons and to the fluxes through the biochemical network [for a review, see reference *4*]. This approach has been used to analyze biochemical pathways in many systems, including hepatocytes (*5, 6*), kidney cortex (*7*), and yeast (*8*). In addition, theoretical treatments of the use of isotopes in the study of the TCA cycle and other biochemical pathways have been performed by many groups (*9–11*). Although some groups have used numerical methods, and others have derived algebraic equations to solve for metabolic fluxes, all of the previous methods of isotope distribution analysis have treated each atom individually. This can result in a large number of equations, which are cumbersome to manipulate and to solve. Furthermore, the resulting equations bear little resemblance to the original metabolic network. Finally, each time the network is modified or refined, the equations must be derived and solved again.

The method of analyzing isotope distributions in metabolic networks proposed in this paper addresses the shortcomings of conventional techniques. In the method described here, the formulation of the biochemical network and the equations describing the steady state isotope distribution are decoupled from the details of the transfer of carbon atoms from reactants to products. The resulting equations are easy to understand, and their solution is obtained iteratively via computer. There is

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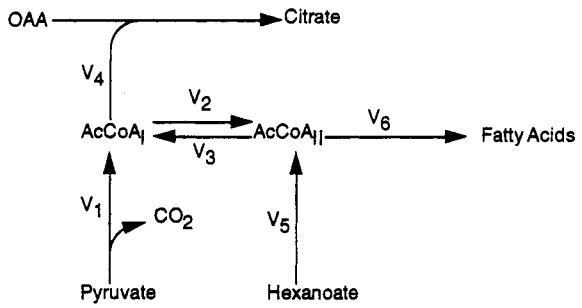

Figure 1. Simple metabolic network for modeling AcCoA metabolism. V_1-V_6 represent fluxes (typical units: millimoles/cell/hour). Pyruvate and hexanoate are potentially labeled substates (based on network analyzed by Blum and Stein (1982)).

virtually no algebra involved, so that modifications to the network are straightforward. The proposed method is an intuitive and flexible way of relating intracellular fluxes to isotope distribution.

In this paper, we first analyze a simple biochemical network using conventional methods. We then present the analysis of fluxes and isotope distributions of the same network using atom mapping matrices. Finally, the proposed method is applied to the analysis of intracellular fluxes in a hybridoma cell line.

2. Conventional Analysis

To illustrate the conventional analysis of isotope distribution and flux estimation in biochemical networks, an example presented by Blum and Stein (*4*) will be used. Consider the simple network shown in Figure 1 representing a two-pool model of AcCoA metabolism. In this system, pyruvate and hexanoate are substrates that can potentially be labeled with ^{13}C or ^{14}C. The distribution of isotope in the atoms of AcCoA in pools I and II will be a function of the carbon fluxes V_1-V_6 (which have units millimoles/cell/hour) and of the level and pattern of substrate labeling.

In order to estimate fluxes, the mathematical relationship between the specific activity (fractional enrichment) of AcCoA carbons and the intracellular fluxes must be determined. An isotopic steady state usually is assumed, so that the flow of label into a specific carbon of an intracellular intermediate is exactly balanced by the flow of labeled carbons out. For the analysis that follows, the specific activity of a particular carbon atom is represented by the metabolite name followed by the carbon atom in parentheses.

Let us construct the steady state balance for the first carbon of AcCoA in pool I (designated $AcCoA_I(1)$). There are two reactions that generate $AcCoA_I$: one is the decarboxylation of pyruvate, which has a total flux of V_1, and the other is the transfer of AcCoA from pool II to pool I, which has a flux of V_3. Carbon 1 of pyruvate is lost to CO_2, while carbons 2 and 3 of pyruvate become carbons 1 and 2, respectively, of $AcCoA_I$. Therefore, the flux of label from pyruvate to carbon 1 of $AcCoA_I$ is given by the product $V_1Pyr(2)$. The conversion of $AcCoA_{II}$ to $AcCoA_I$ does not involve a chemical reaction, so that carbons 1 and 2 of $AcCoA_{II}$ become carbons 1 and 2, respectively, of $AcCoA_I$. Thus, the flux of label from $AcCoA_{II}$ to carbon 1 of $AcCoA_I$ is given by the product $V_3AcCoA_{II}(1)$. The total flux of label into $AcCoA_I(1)$ is the sum of the two individual fluxes:

$$\text{flux into } AcCoA_I(1) = V_1Pyr(2) + V_3AcCoA_{II}(1) \quad (1)$$

The two reactions that remove $AcCoA_I$ are the condensation with oxaloacetate (OAA) to form citrate, which has a flux of V_4, and the transfer of AcCoA from pool I to pool II, which has a flux of V_2. The flux of label out of carbon 1 of $AcCoA_I$ is simply $AcCoA_I(1)$ multiplied by the sum of the two fluxes, V_2 and V_4:

$$\text{flux out of } AcCoA_I(1) = (V_2 + V_4)AcCoA_I(1) \quad (2)$$

At steady state the flux of label into $AcCoA_I(1)$ will be exactly balanced by the flux out. Equating eqs 1 and 2 yields the steady state balance of label for carbon 1 of $AcCoA_I$:

$$(V_2 + V_4)AcCoA_I(1) = V_1Pyr(2) + V_3AcCoA_{II}(1) \quad (3)$$

Similarly, the steady state isotope balance for carbon 2 of $AcCoA_I$ is given by

$$(V_2 + V_4)AcCoA_I(2) = V_1Pyr(3) + V_3AcCoA_{II}(2) \quad (4)$$

The derivation of the steady state equations can be repeated for AcCoA in pool II. There are two reactions that generate $AcCoA_{II}$: one is the oxidation of hexanoate, which has a total flux of V_5, and the other is the transfer of AcCoA from pool I to pool II, which has a flux of V_2. The six carbons of hexanoate are removed in pairs via β-oxidation, so that the odd and even carbons each contribute one-third of the specific activity to carbons 1 and 2 of $AcCoA_{II}$, respectively. Therefore, the flux of label from hexanoate to carbon 1 of $AcCoA_{II}$ is given by the product $(V_5/3)(Hex(1) + Hex(3) + Hex(5))$. The conversion of $AcCoA_{II}$ to $AcCoA_I$ does not involve a chemical reaction, so that the flux of label from $AcCoA_I$ to carbon 1 of $AcCoA_{II}$ is given by the product $V_2AcCoA_I(1)$. The total flux of label into $AcCoA_{II}(1)$ is the sum of the two individual fluxes:

$$\text{flux into } AcCoA_{II}(1) = (V_5/3)(Hex(1) + Hex(3) + Hex(5)) + V_2AcCoA_I(1) \quad (5)$$

The flux of label out of carbon 1 of $AcCoA_{II}$ is the specific activity $AcCoA_{II}(1)$ multiplied by the sum of the two fluxes, V_3 and V_6:

$$\text{flux out of } AcCoA_{II}(1) = (V_3 + V_6)AcCoA_{II}(1) \quad (6)$$

Equating eqs 5 and 6 yields the steady state balance of label for carbon 1 of $AcCoA_{II}$:

$$(V_3 + V_6)AcCoA_{II}(1) = (V_5/3)(Hex(1) + Hex(3) + Hex(5)) + V_2AcCoA_I(1) \quad (7)$$

Similarly, the steady state isotope balance for carbon 2 of $AcCoA_{II}$ is given by

$$(V_3 + V_6)AcCoA_{II}(2) = (V_5/3)(Hex(2) + Hex(4) + Hex(6)) + V_2AcCoA_I(2) \quad (8)$$

Equations 3, 4, 7, and 8 relate the fluxes through the network, V_1-V_6, and the specific activities of the substrate carbon atoms, Pyr(1−3) and Hex(1−6), to the specific activities of the intermediate carbon atoms, $AcCoA_I(1,2)$ and $AcCoA_{II}(1,2)$. For small networks like this one, the system of equations can often be solved to give explicit expresions for the fluxes (or, frequently, flux ratios) in terms of the specific activities of the intermediates. However, for larger, more complicated biochemical networks, it is difficult or impossible to obtain explicit expressions for the fluxes. Instead, the specific activities

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usually are found as a function of the intracellular fluxes. The set of fluxes that best fits the observed isotope distribution is then determined by trial and error.

One method for determining the specific activities of intermediates is algebraic manipulation of the steady state equations. For example, eqs 3, 4, 7, and 8 can be combined to find an explicit expression for the specific activity of carbon 1 of $AcCoA_I$:

$$AcCoA_I(1) = \frac{V_1V_2}{V_2V_6 + V_3V_4 + V_4V_6}Pyr(2) + \frac{V_5(V_2 + V_4)}{3(V_2V_6 + V_3V_4 + V_4V_6)}(Hex(1) + Hex(3) + Hex(5)) \quad (9)$$

Similar expressions can be derived for $AcCoA_I(2)$, $AcCoA_{II}(1)$, and $AcCoA_{II}(2)$. The explicit solution for specific activities is practical for small networks although, as evidenced by eq 9, even simple biochemical networks can result in cumbersome equations. In addition, modifications of the network require rederivation of the equations.

Another procedure for solving for the specific activities is to arrange the steady state equations into matrix form and solve by matrix inversion:

$$\mathbf{As} = \mathbf{b} \quad (10)$$

In this equation, **s** is the vector of the specific activities of the intermediate carbon atoms (unknowns), **b** is the vector of the specific activities of the substrates (knowns), and **A** is a matrix whose elements are the coefficients of the steady state equations (which are, in turn, combinations of the fluxes). Returning to the example network in Figure 1, we can write eqs 3, 4, 7, and 8 in matrix form as

$$\begin{bmatrix} \frac{V_2+V_4}{V_1} & 0 & -\frac{V_3}{V_1} & 0 \\ 0 & \frac{V_2+V_4}{V_1} & 0 & -\frac{V_3}{V_1} \\ -\frac{3V_2}{V_5} & 0 & \frac{3(V_3+V_6)}{V_5} & 0 \\ 0 & -\frac{3V_2}{V_5} & 0 & \frac{3(V_3+V_6)}{V_5} \end{bmatrix} \times \begin{bmatrix} AcCoA_I(1) \\ AcCoA_I(2) \\ AcCoA_{II}(1) \\ AcCoA_{II}(2) \end{bmatrix} = \begin{bmatrix} Pyr(2) \\ Pyr(3) \\ Hex(1)+Hex(3)+Hex(5) \\ Hex(2)+Hex(4)+Hex(6) \end{bmatrix} \quad (11)$$

The procedure for solving this equation for the specific activities in $AcCoA_I$ and $AcCoA_{II}$ is to choose a set of fluxes to generate the **A** matrix and then solve for **s** by inverting **A**:

$$\mathbf{s} = \mathbf{A}^{-1}\mathbf{b} \quad (12)$$

While this method is a significant improvement over the algebraic solution, there are still a fair number of manipulations that must be performed, and modifications to the network may result in laborious revisions to the equations. Furthermore, the dimensions of the **A** matrix can become quite large in a detailed metabolic network, requiring significant computing power.

3. Atom Mapping Matrices

An alternative method of modeling isotope distributions avoids several of the shortcomings of the conventional analysis. In this method, atom mapping matrices (AMMs) are introduced describing the transfer of carbon atoms from reactants to products. The AMMs decouple the formulation of the biochemical network and the steady state equations governing isotope distribution from the details of the reaction mechanisms. The resulting equations are easy to derive, modify, and solve. The method using AMMs will be introduced using the simple biochemical network in Figure 1 and will then be applied to the analysis of intracellular fluxes in a hybridoma cell line.

3.1. Application to Example Network. The first step in this method is to represent the specific activities of the carbon atoms of each metabolite in vector form. The *n*th element of a metabolite vector contains the specific activity of the *n*th carbon. For the reaction network in Figure 1, the following metabolite vectors are formed:

$$\mathbf{Pyr} = \begin{bmatrix} Pyr(1) \\ Pyr(2) \\ Pyr(3) \end{bmatrix} \quad \mathbf{AcCoA_I} = \begin{bmatrix} AcCoA_I(1) \\ AcCoA_I(2) \end{bmatrix}$$

$$\mathbf{Hex} = \begin{bmatrix} Hex(1) \\ Hex(2) \\ Hex(3) \\ Hex(4) \\ Hex(5) \\ Hex(6) \end{bmatrix} \quad \mathbf{AcCoA_{II}} = \begin{bmatrix} AcCoA_{II}(1) \\ AcCoA_{II}(2) \end{bmatrix}$$

With vector notation, all of the carbon atoms of a metabolite are represented in a clear and concise form.

The next step is to construct atom mapping matrices for the reactions in the metabolic network. These matrices describe the transfer of atoms from reactants to products. For each reaction there will be mapping matrices for every reactant−product pair. For example, consider a general reaction involving two reactants, A and B, and two products, C and D, catalyzed by enzyme E:

$$A + B \xrightarrow{E} C + D \quad (13)$$

This reaction will have four mapping matrices describing the transfer of carbons from A to C, A to D, B to C, and B to D. We designate the mapping matrices for each reaction by square brackets followed by the subscripted enzyme (or reaction) name. Inside the square brackets the particular reactant−product pair is indicated, separated by a > (to indicate direction). Thus, the four mapping matrices for the sample reaction are as follows:

$[\mathbf{A>C}]_\mathbf{E}$ describes transfer of carbon from A to C

$[\mathbf{A>D}]_\mathbf{E}$ describes transfer of carbon from A to D

$[\mathbf{B>C}]_\mathbf{E}$ describes transfer of carbon from B to C

$[\mathbf{B>D}]_\mathbf{E}$ describes transfer of carbon from B to D

The atom mapping matrices are constructed such that multiplication of the reactant specific activity vector by the AMM specifies the contribution to the product specific activity vector. The specific activity of the resulting

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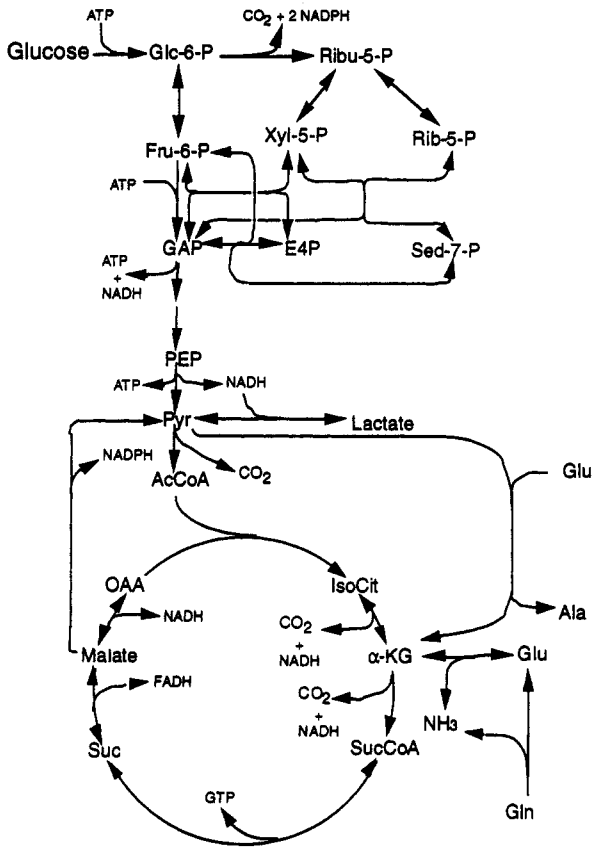

Figure 2. Biochemistry of energy metabolism in hybridomas. The energy metabolism of mammalian cells consists of glycolysis, glutaminolysis, the pentose phosphate pathway, and the TCA cycle working together to produce ATP, NADPH, and biosynthetic precursors.

products will be the sum of the contribution from each reactant:

$$[\mathbf{A}>\mathbf{C}]_{\mathbf{E}}\mathbf{A} + [\mathbf{B}>\mathbf{C}]_{\mathbf{E}}\mathbf{B} = \mathbf{C} \tag{14}$$

$$[\mathbf{A}>\mathbf{D}]_{\mathbf{E}}\mathbf{A} + [\mathbf{B}>\mathbf{D}]_{\mathbf{E}}\mathbf{B} = \mathbf{D} \tag{15}$$

The dimensions of the mapping matrices are determined by the number of carbons in the reactant and product. The number of columns equals the number of atoms in the reactant, while the number of rows equals the number of carbons in the product. The element in the ith row and the jth column of the mapping matrix specifies the amount of the ith carbon of the product that is derived from the jth carbon of the reactant. Typically, there is a definite and unique mapping of reactant carbons to product carbons, so that the elements of the mapping matrix are usually 0 or 1. However, fractional elements are possible.

To return to the sample metabolic network, let us construct the mapping matrices. Reaction 1 is catalyzed by pyruvate dehydrogenase (PD) with pyruvate as the only reactant and $AcCoA_I$ and CO_2 as the two products. In this analysis we have not included CO_2 as a measurable metabolite, so that only one mapping matrix is needed: $[\mathbf{Pyr}>\mathbf{AcCoA_I}]_{\mathbf{PD}}$. We know that carbon 1 of pyruvate is lost to CO_2, and carbons 2 and 3 go to carbons 1 and 2, respectively, of $AcCoA_I$. This results in the following atom mapping matrix:

$$[\mathbf{Pyr}>\mathbf{AcCoA_I}]_{\mathbf{PD}} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{16}$$

If $AcCoA_I$ were derived only from pyruvate, then its

Table 1. Set of Reactions Describing Energy Metabolism in CRL-1606 Hybridoma[a]

1:	glc + ATP → glc-6-P + ADP
2:	glc-6-P → fru-6-P
3:	fru-6-P + ATP → 2 GAP + ADP
4:	GAP + 2 ADP + NAD → pyruvate + NADH + 2 ATP
5:	pyruvate + NADH → lactate + NAD
6:	glc-6-P + 2 NADP → ribulose-5-P + 2 NADPH + CO_2
7:	ribulose-5-P ↔ xylulose-5-P
8:	ribulose-5-P + xylulose-5-P ↔ sed-7-P + GAP
9:	sed-7-P + GAP ↔ fru-6-P + erythrose-4-P
10:	xylulose-5-P + erythrose-4-P ↔ fru-6-P + GAP
11:	pyruvate + malate + 3 NAD → α-KG + 3 NADH + 2 CO_2
12:	a-KG + 2 NAD + ADP → 2 NADH + CO_2 + ATP + malate
13:	glutamine → glutamate + NH_3
14:	pyruvate + glutamate → alanine + α-KG
15:	glutamate + NAD ↔ α-KG + NADH + NH_3
16:	2 NADH + 6 ADP + O_2 → 2 NAD + 6 ATP
17:	malate + NADP → pyruvate + CO_2 + NADPH
18:	ATP → ADP
19:	0.036 glutamine + 0.062 glutamate + 0.0031 glc-6-P + 0.013 ribulose-5-P + 0.042 GAP + 0.123 pyruvate + 0.019 malate + 0.74 ATP + 0.188 NADPH + 0.23 NAD → biomass + 0.073 a-KG + 0.23 NADH + 0.12 CO_2 + 0.74 ADP + 0.188 NADP
20:	0.012 glutamine + 0.07 glutamate + 0.04 GAP + 0.011 NADPH + 0.012 malate + 0.082 NAD → Ab + 0.082 NADH + 0.057 α-KG + 0.011 NADP

[a] This represents the network presented in Figure 2, with some intermediates that are only involved in two reactions removed.

specific activity would be given by

$$[\mathbf{Pyr}>\mathbf{AcCoA_I}]_{\mathbf{PD}}\mathbf{Pyr} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} \mathrm{Pyr}(1) \\ \mathrm{Pyr}(2) \\ \mathrm{Pyr}(3) \end{bmatrix} = \begin{bmatrix} \mathrm{Pyr}(2) \\ \mathrm{Pyr}(3) \end{bmatrix} = \begin{bmatrix} \mathrm{AcCoA_I}(1) \\ \mathrm{AcCoA_I}(2) \end{bmatrix} = \mathbf{AcCoA_I} \tag{17}$$

However, there is also transfer of AcCoA between pools I and II, so that eq 17 does not completely determine the specific activity of $AcCoA_I$.

Reaction 2 is the transport of AcCoA from pool I to pool II and is designated by the name transI, while reaction 3 is the transport of AcCoA from pool II to pool I and is named transII. Since the carbons in $AcCoA_I$ and $AcCoA_{II}$ map directly to each other, $[\mathbf{AcCoA_I}>\mathbf{AcCoA_{II}}]_{\mathbf{transI}}$ and $[\mathbf{AcCoA_{II}}>\mathbf{AcCoA_I}]_{\mathbf{transII}}$ are both 2 × 2 identity matrices:

$$[\mathbf{AcCoA_I}>\mathbf{AcCoA_{II}}]_{\mathbf{transI}} = [\mathbf{AcCoA_{II}}>\mathbf{AcCoA_I}]_{\mathbf{transII}} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

The final reaction for which there is an atom mapping matrix is reaction 5, the oxidation of hexanoate to $AcCoA_{II}$, and is designated βox. The β-oxidation of hexanoate removes carbons in pairs to form three AcCoA molecules. The odd carbons of hexanoate are indistinguishable upon conversion to $AcCoA_{II}$, so that their coefficients in the matrix $[\mathbf{Hex}>\mathbf{AcCoA_{II}}]_{\beta\mathbf{ox}}$ are one-third, and the same holds true for the even carbons. The odd and even carbons are mapped to carbons 1 and 2, respectively, of $AcCoA_{II}$, giving

$$[\mathbf{Hex}>\mathbf{AcCoA_{II}}]\beta_{\mathbf{ox}} = \begin{bmatrix} \frac{1}{3} & 0 & \frac{1}{3} & 0 & \frac{1}{3} & 0 \\ 0 & \frac{1}{3} & 0 & \frac{1}{3} & 0 & \frac{1}{3} \end{bmatrix} \tag{18}$$

All of the necessary atom mapping matrices have been constructed and can be used to formulate the steady state

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isotope balance equations. The flux of label into a metabolite is simply the sum of the products of the mapping matrices and the reactant specific activity vectors, weighted by the corresponding reaction flux. For $AcCoA_I$, the two contributing reactions are PD with flux V_1 and transII with flux rate V_2:

$$\text{flux into } \mathbf{AcCoA_I} = V_1[\mathbf{Pyr}>\mathbf{AcCoA_I}]_{\mathbf{PD}}\mathbf{Pyr} + V_3[\mathbf{AcCoA_{II}}>\mathbf{AcCoA_I}]_{\mathbf{transII}}\mathbf{AcCoA_{II}} \quad (19)$$

The flux of label out of $AcCoA_I$ is

$$\text{flux out of } \mathbf{AcCoA_I} = (V_2 + V_4)\mathbf{AcCoA_I} \quad (20)$$

Equating eqs 19 and 20 gives the steady state isotope balance for $AcCoA_I$:

$$(V_2 + V_4)\mathbf{AcCoA_I} = V_1[\mathbf{Pyr}>\mathbf{AcCoA_I}]_{\mathbf{PD}}\mathbf{Pyr} + V_2[\mathbf{AcCoA_{II}}>\mathbf{AcCoA_I}]_{\mathbf{transII}}\mathbf{AcCoA_{II}} \quad (21)$$

Similarly, the steady state isotope balance for $AcCoA_{II}$ is

$$(V_3 + V_6)\mathbf{AcCoA_{II}} = V_2[\mathbf{AcCoA_I}>\mathbf{AcCoA_{II}}]_{\mathbf{transI}}\mathbf{AcCoA_I} + V_5[\mathbf{Hex}>\mathbf{AcCoA_I}]_{\beta\mathbf{ox}}\mathbf{Hex} \quad (22)$$

Equations 21 and 22 are equivalent to eqs 3, 4, 7, and 8 obtained by treating each atom separately. In a small network like this example, the level of complexity of the two methods is similar. However, for larger networks, especially those with many metabolites possessing more than three carbons, construction of atom mapping matrices and formulation of the steady state balances in matrix form result in a more representative and compact system of equations. In addition, if new information about the way the carbons are transferred from reactant to product becomes available, only the atom mapping matrices describing the affected reaction need to be altered. This is rather straightforward and requires no new algebra.

The set of equations describing isotope distributions in a metabolic network using atom mapping matrices can be solved iteratively via computer. This requires that the specific activities of the substrate carbon atoms be initialized (to values between 0 and 1 for fractional enrichment) and a consistent set of flux values be provided. Then, each of the steady state equations is solved sequentially for the activities of the output metabolites, and the process is repeated until convergence is achieved. This is essentially equivalent to solving $\mathbf{As} = \mathbf{b}$ using the Gauss–Seidel method, with the equation partitioned into smaller matrix equations. The relationship between the iterative solution of the vector equations with AMMs, and the Gauss–Seidel method applied to the single matrix equation, is detailed in Appendix A. All of the matrices are small and no matrix inversions are necessary, so that this method is not computationally demanding, even for very large biochemical networks.

3.2. Application to Hybridoma Metabolism. In the example presented in Figure 1, the algebraic effort involved in the standard analysis was not substantial, but in practice, metabolic networks are likely to be much larger and more complex. To demonstrate the ease and clarity of the mapping matrix method, we applied it to the energy metabolism of hybridoma cells. The biochemical network of energy metabolism in a hybridoma cell line (ATTC CRL-1606) is presented in Figure 2. The stoichiometric reactions represented in Figure 2 are shown in Table 1. Reactions 19 and 20 represent the syntheses of biomass and antibody, respectively. The stoichiometry for biomass was estimated from the macromolecular composition of CRL-1606 cells and the known biosynthetic pathways (*12*). The equation for antibody was derived in a similar fashion. These two reactions are included for completeness (to account for the drain of metabolites for biosynthesis), but are not, strictly speaking, involved in energy metabolism. The metabolite vectors for this network are compiled in Table 2.

Table 2. Definitions of the Metabolite Specific Activity Vectors for the Hybridoma Network in Figure 2

Erythrose-4-P: $\mathbf{e4p} = \begin{bmatrix} e4p(1) \\ e4p(2) \\ e4p(3) \\ e4p(4) \end{bmatrix}$

α-Ketoglutarate: $\mathbf{aKG} = \begin{bmatrix} aKG(1) \\ aKG(2) \\ aKG(3) \\ aKG(4) \\ aKG(5) \end{bmatrix}$

Fructose-6-P: $\mathbf{f6p} = \begin{bmatrix} f6p(1) \\ f6p(2) \\ f6p(3) \\ f6p(4) \\ f6p(5) \\ f6p(6) \end{bmatrix}$

Lactate: $\mathbf{lac} = \begin{bmatrix} lac(1) \\ lac(2) \\ lac(3) \end{bmatrix}$

Malate: $\mathbf{mal} = \begin{bmatrix} mal(1) \\ mal(2) \\ mal(3) \\ mal(4) \\ mal(5) \end{bmatrix}$

Glucose: $\mathbf{glc} = \begin{bmatrix} glc(1) \\ glc(2) \\ glc(3) \\ glc(4) \\ glc(5) \\ glc(6) \end{bmatrix}$

Pyruvate: $\mathbf{pyr} = \begin{bmatrix} pyr(1) \\ pyr(2) \\ pyr(3) \end{bmatrix}$

Glucose-6-P: $\mathbf{g6p} = \begin{bmatrix} g6p(1) \\ g6p(2) \\ g6p(3) \\ g6p(4) \\ g6p(5) \\ g6p(6) \end{bmatrix}$

Ribulose-5-P: $\mathbf{r5p} = \begin{bmatrix} r5p(1) \\ r5p(2) \\ r5p(3) \\ r5p(4) \\ r5p(5) \end{bmatrix}$

Glutamine: $\mathbf{gln} = \begin{bmatrix} gln(1) \\ gln(2) \\ gln(3) \\ gln(4) \\ gln(5) \end{bmatrix}$

Sedoheptulose-7-P: $\mathbf{s7p} = \begin{bmatrix} s7p(1) \\ s7p(2) \\ s7p(3) \\ s7p(4) \\ s7p(5) \\ s7p(6) \\ s7p(7) \end{bmatrix}$

Glyceraldehyde-3-P: $\mathbf{gap} = \begin{bmatrix} gap(1) \\ gap(2) \\ gap(3) \end{bmatrix}$

Xylulose-5-P: $\mathbf{x5p} = \begin{bmatrix} x5p(1) \\ x5p(2) \\ x5p(3) \\ x5p(4) \\ x5p(5) \end{bmatrix}$

The atom mapping matrices for the hybridoma network are presented in Table 3. To simplify the nomenclature, the reaction number (from Table 1), rather than the catalyzing enzyme, will be used as the subscript in the AMM. For the reverse reactions (in this case, the "gluconeogenic" reactions), an r is included in the subscript. To further clarify the method, a few selected mapping matrices will be considered here in detail. The matrices $[\mathbf{glc}>\mathbf{g6p}]_1$, $[\mathbf{f6p}>\mathbf{g6p}]_{2r}$, and $[\mathbf{g6p}>\mathbf{f6p}]_2$ describe the interconversion of the six-carbon sugars, glucose, glucose 6-phosphate (glucose-6-P), and fructose-6-phosphate (fructose-6-P). Since the carbon atoms retain their relative positions, the three matrices equal the 6 × 6 identity matrix:

$$[\mathbf{glc}>\mathbf{g6p}]_1 = [\mathbf{f6p}>\mathbf{g6p}]_{2r} = [\mathbf{g6p}>\mathbf{f6p}]_2 = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \quad (23)$$

The matrices $[\mathbf{x5p}>\mathbf{f6p}]_{10}$ and $[\mathbf{e4p}>\mathbf{f6p}]_{10}$ map carbons from xylulose 5-phosphate (xylulose-5-P) and erythrose 4-phosphate (erythrose-4-P) to fructose-6-P. Xylulose-5-P

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is a five-carbon sugar whose 1 and 2 carbons become the 1 and 2 carbons of fructose-5-P when combined with erythrose-4-P in the reaction catalyzed by a transketolase. The matrix for this mapping is

$$[\mathbf{x5p>f6p}]_{10} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \tag{24}$$

The remaining carbons 3–6 of fructose-6-P come from carbons 1–4 of erythrose-4-P, resulting in the following mapping matrix:

$$[\mathbf{e4p>f6p}]_{10} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{25}$$

In the matrices considered so far, all of the elements were either 0 or 1, but there are two situations that give rise to fractional matrix elements. As was shown previously in the AcCoA example, a molecule that reacts to become two or more of another species will have fractional elements. In this reaction network, fructose-6-P is converted into two molecules of GAP, and the corresponding mapping matrix contains fractional elements as follows:

$$[\mathbf{f6p>GAP}]_3 = \begin{bmatrix} 0 & 0 & \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ 0 & \frac{1}{2} & 0 & 0 & \frac{1}{2} & 0 \\ \frac{1}{2} & 0 & 0 & 0 & 0 & \frac{1}{2} \end{bmatrix} \tag{26}$$

Another sitution that results in fractional elements involves the generation of symmetric molecules. In the TCA cycle, α-ketoglutarate is converted to malate, with succinate and fumarate as intermediates. Both succinate and fumarate are symmetrical, so that when fumarate is converted to malate, the 1 and 4 carbons will have the same labeling, as will the 2 and 3 carbons. The mapping matrix that accounts for this scrambling of label is

$$[\mathbf{aKG>mal}]_{12} = \begin{bmatrix} 0 & \frac{1}{2} & 0 & 0 & \frac{1}{2} \\ 0 & 0 & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & \frac{1}{2} & 0 & 0 & \frac{1}{2} \end{bmatrix} \tag{27}$$

Finally, some mapping matrices will consist of all zeroes when none of the source carbons contribute to a particular product in a reaction. For example, in the transamination reaction,

glutamate + pyruvate → α-ketoglutarate + alanine

none of the pyruvate carbon is incorporated into α-ketoglutarate, so that the mapping matrix consists of all zeroes:

$$[\mathbf{pyr>aKG}]_{14} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \tag{28}$$

These examples demonstrate the typical types of

Table 3. Definitions of the Atom Mapping Matrices for the Hybridoma Network in Figure 2

$$[\mathbf{glc>g6p}]_1 = [\mathbf{f6p>g6p}]_{2r} = [\mathbf{g6p>f6p}]_2 = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$[\mathbf{x5p>f6p}]_{10} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \qquad [\mathbf{f6p>gap}]_3 = \begin{bmatrix} 0 & 0 & \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ 0 & \frac{1}{2} & 0 & 0 & \frac{1}{2} & 0 \\ \frac{1}{2} & 0 & 0 & 0 & 0 & \frac{1}{2} \end{bmatrix}$$

$$[\mathbf{e4p>f6p}]_{10} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \qquad [\mathbf{x5p>gap}]_{10} = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$[\mathbf{e4p>gap}]_{10} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$[\mathbf{s7p>f6p}]_9 = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \qquad [\mathbf{r5p>gap}]_8 = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$[\mathbf{x5p>gap}]_8 = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$[\mathbf{gap>f6p}]_9 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \qquad [\mathbf{mal>pyr}]_{17} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$[\mathbf{pyr>aKG}]_{11} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$[\mathbf{gap>f6p}]_{3r} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$[\mathbf{mal>aKG}]_{11} = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \qquad [\mathbf{r5p>s7p}]_8 = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$[\mathbf{pyr>gap}]_{4r} = [\mathbf{gap>pyr}]_4 = [\mathbf{pyr>lac}]_5 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$[\mathbf{gln>aKG}]_{15} = [\mathbf{gln>aKG}]_{14} = [\mathbf{r5p>x5p}]_7 = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$[\mathbf{pyr>aKG}]_{14} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \qquad [\mathbf{s7p>e4p}]_9 = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$[\mathbf{aKG>mal}]_{12} = \begin{bmatrix} 0 & \frac{1}{2} & 0 & 0 & \frac{1}{2} \\ 0 & 0 & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & \frac{1}{2} & 0 & 0 & \frac{1}{2} \end{bmatrix} \qquad [\mathbf{gap>e4p}]_9 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$[\mathbf{x5p>s7p}]_8 = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$[\mathbf{g6p>r5p}]_6 = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

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Table 4. Matrix Equations Governing the Flow of Isotope Through the Metabolic Network at Isotopic Steady State[a]

$$(x_6 + 0.0031\, x_{19} + x_2)\ \mathbf{g6p} = x1\ [\mathbf{glc{>}g6p}]_1\ \mathbf{glc} + x_{2r}\ [\mathbf{f6p{>}g6p}]_{2r}\ \mathbf{f6p}$$

$$(x_{2r} + x_3)\ \mathbf{f6p} = x_2\ [\mathbf{g6p{>}f6p}]_2\ \mathbf{g6p} + x_{10}\,(\,[\mathbf{x5p{>}f6p}]_{10}\ \mathbf{x5p} + [\mathbf{e4p{>}f6p}]_{10}\ \mathbf{e4p}) + x_9\,(\,[\mathbf{s7p{>}f6p}]_9\ \mathbf{s7p} + [\mathbf{gap{>}f6p}]_9\ \mathbf{gap}) + x_{3r}\ [\mathbf{gap{>}f6p}]_{3r}\ \mathbf{gap}$$

$$(x_4 + 0.042\, x_{19} + x_9 + 2\, x_{3r})\ \mathbf{gap} = 2\, x_3\ [\mathbf{f6p{>}gap}]_3\ \mathbf{f6p} + x_{10}\,(\,[\mathbf{x5p{>}gap}]_{10}\ \mathbf{x5p} + [\mathbf{e4p{>}gap}]_{10}\ \mathbf{e4p}) + x_8\,(\,[\mathbf{r5p{>}gap}]_8\ \mathbf{r5p} + [\mathbf{x5p{>}gap}]_8\ \mathbf{x5p}) + x_{4r}\ [\mathbf{pyr{>}gap}]_{4r}\ \mathbf{pyr}$$

$$(x_{11} + x_5 + x_{4r} + 0.123\, x_{19} + x_{14})\ \mathbf{pyr} = x_4\ [\mathbf{gap{>}pyr}]_4\ \mathbf{gap} + x_{17}\ [\mathbf{mal{>}pyr}]_{17}\ \mathbf{mal}$$

$$x_5\ \mathbf{lac} = x_5\ [\mathbf{pyr{>}lac}]_5\ \mathbf{pyr}$$

$$x_{12}\ \mathbf{aKG} = x_{11}\,(\,[\mathbf{pyr{>}aKG}]_{11}\ \mathbf{pyr} + [\mathbf{mal{>}aKG}]_{11}\ \mathbf{mal}) + (x_{15} + 0.073\, x_{19})\ [\mathbf{gln{>}aKG}]_{15}\ \mathbf{gln} + x_{14}\,(\,[\mathbf{pyr{>}aKG}]_{14}\ \mathbf{pyr} + [\mathbf{gln{>}aKG}]_{14}\ \mathbf{gln})$$

$$(x_{11} + x_{17} + 0.019\, x_{19})\ \mathbf{mal} = x_{12}\ [\mathbf{aKG{>}mal}]_{12}\ \mathbf{aKG}$$

$$(x_7 + x_8 + 0.013\, x_{19})\ \mathbf{r5p} = x_6\ [\mathbf{g6p{>}r5p}]_6\ \mathbf{g6p}$$

$$(x_8 + x_{10})\ \mathbf{x5p} = x_7\ [\mathbf{r5p{>}x5p}]_7\ \mathbf{r5p}$$

$$x_{10}\ \mathbf{e4p} = x_9\,(\,[\mathbf{s7p{>}e4p}]_9\ \mathbf{s7p} + [\mathbf{gap{>}e4p}]_9\ \mathbf{gap})$$

$$x_9\ \mathbf{s7p} = x_8\,(\,[\mathbf{x5p{>}s7p}]_8\ \mathbf{x5p} + [\mathbf{r5p{>}s7p}]_8\ \mathbf{r5p})$$

[a] Equations are derived from reaction stoichiometry shown in Table 1. The symbols x_i and x_{ir} denote the forward and reverse fluxes, respectively, through reaction i (from Table 1).

mapping matrices that result from biochemical reactions. They are easy to generate individually and are not complicated by the presence of other reactions.

The steady state isotope balance equations are constructed by balancing the flow of label into a metabolite with the flow of label out. The equations are derived from the stoichiometry presented in Table 1. In practice, one simply constructs a total mass balance around each metabolite and then inserts the mapping matrices and the metabolite specific activity vectors. The forward intracellular fluxes are designated by x_i and reverse fluxes by x_{ir}, where i is the number of the reaction (from Table 1). Typical units for the flux through a biochemical pathway are moles/cell/hour. To get the flux into, or out of, a metabolite, the reaction flux, x_i, must be multiplied by the metabolite's stoichiometric coefficient. For example, glucose-6-P is derived from glucose in reaction 1 and fructose-6-P in the reverse of reaction 2. It is consumed in reaction 2 to form fructose-6-P, in reaction 6 to form ribulose-5-P, and in reaction 19 to form biomass. A mass balance around glucose-6-P gives the following equation:

$$(x_2 + x_6 + 0.0031x_{19}) = x_1 + x_{2r} \tag{29}$$

Insertion of the mapping matrices and metabolite vectors transforms it into an isotope balance:

$$(x_2 + x_6 + 0.0031x_{19})\mathbf{g6p} = x_1[\mathbf{glc{>}g6p}]_1\mathbf{glc} + x_2[\mathbf{f6p{>}g6p}]_2\mathbf{f6p} \tag{30}$$

Using the same procedure, the remaining steady state equations were constructed and are presented in Table 4.

To demonstrate how the modeling of isotope distribution can be applied to the analysis of intracellular fluxes, we will discuss its application to the estimation of TCA cycle fluxes with *in vitro* ^{13}C NMR. In CRL-1606 hybridomas, the contribution of the pentose phosphate pathway is expected to be small (<5% that of glycolysis); thus, for the purposes of this analysis, it is neglected. The "gluconeogenic" reactions (2r, 3r, and 4r) are also neglected because hybridomas have very high rates of glycolysis. In addition, the production of antibody and biomass primarily consumes metabolites, and it is only the formation of α-ketoglutarate as a byproduct that contributes to the labeling of TCA cycle intermediates. All of the AMMs for the production of α-ketoglutarate from glutamate are the same (the identity matrix), so that all of the fluxes into α-ketoglutarate are lumped together into a single flux: $x_G = x_{14} + x_{15} + 0.073x_{19} + 0.057x_{20}$. The simplified biochemical network is shown in Figure 3. There are only two branch points in the network, at pyruvate and α-ketoglutarate, so that the distribution of label throughout the TCA cycle intermediates will be a function of the substrate labeling pattern and two flux ratios. One flux ratio, $f_1 = x_4/(x_4 + x_{17})$, is the fraction of pyruvate derived from glucose via glycolysis. The other ratio, $f_2 = x_G/(x_G + x_{11})$, is the fraction of α-ketoglutarate derived from glutamine via glutamate.

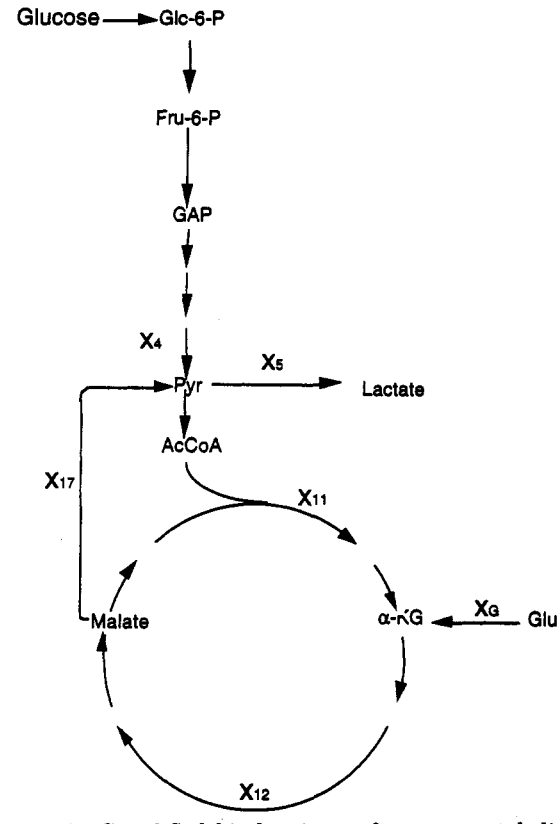

Figure 3. Simplified biochemistry of energy metabolism in hybridomas. The pentose phosphate pathway and "gluconeogenic" (reverse) reactions are neglected. In addition, the fluxes into α-ketoglutarate (x_{14}, x_{15}, x_{19}, and x_{20}) are lumped into a single flux, x_G. The distribution of isotope is a function of the labeling of glucose and glutamine and of the fluxes x_4, x_{11}, x_{17}, and x_G. The isotope distribution in secreted lactate can be measured via NMR.

The complete Mathematica program, including mapping matrix definitions, for modeling the simplified hybridoma metabolism is included in Appendix B (supplementary material). As defined, the network converges rapidly (in less than 10 iteractions). Furthermore, the largest matrix dimension is 7, compared to the more than 60 that would result if the conventional method were used.

Using the network for the hybridoma derived above, the isotope distribution in secreted lactate was determined as a function of f_1 and f_2, assuming glucose labeled at position 1 with a fractional enrichment of 0.1. This would correspond to a typical experimental situation where 0.5 g of 100% [^{13}C]glucose is added to 4.5 g of unlabeled glucose. The other carbons of glucose, and the

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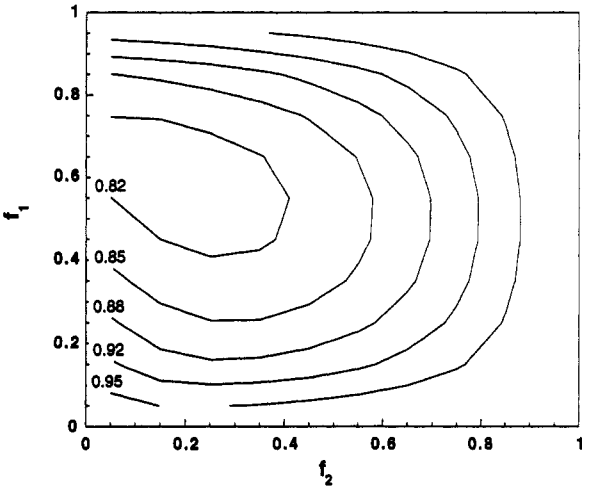

Figure 4. Lactate C1:C2 ratio predicted from f_1 and f_2. The contour lines are values of constant ^{13}C ratios of the carbons in secreted lactate. The simulation assumed that carbon 1 of glucose had a fractional enrichment of 0.1, while all other carbons were at natural abundance levels of 0.011. In addition, it was assumed that 10% of the lactate was not enriched. The isotope distribution in lactate will be a function of the two flux ratios, f_1 and f_2. The flux ratio f_1 is the fraction of pyruvate that is produced from glycolysis, and f_2 is the fraction of α-ketoglutarate derived from glutamate.

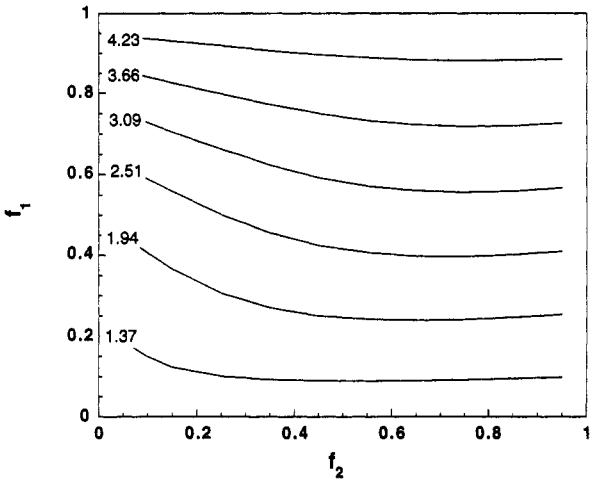

Figure 5. Lactate C3:C1 ratio predicted from f_1 and f_2. The contour lines are values of constant ^{13}C ratios of the carbons in secreted lactate. The simulation assumed that carbon 1 of glucose had a fractional enrichment of 0.1, while all other carbons were at natural abundance levels of 0.011. In addition, it was assumed that 10% of the lactate was not enriched. The isotope distribution in lactate will be a function of the two flux ratios, f_1 and f_2. The flux ratio f_1 is the fraction of pyruvate that is produced from glycolysis, and f_2 is the fraction of α-ketoglutarate derived from glutamate.

carbons of glutamine (the other major substrate), were assumed to be at the natural abundance level of 0.011. The relative enrichment of ^{13}C in the three carbons of lactate was calculated as a function of f_1 and f_2, which can vary from 0 to 1. The presence of unlabeled lactate is unavoidable in practice and was taken to be 10% of the total lactate in this analysis. Relative enrichments (i.e., ratio of label in one position to that in another) are used as these are readily obtained via ^{13}C NMR. The results were expressed in two contour plots: one for the ratio of the amount of label in C1 to that in C2 (C1:C2) and the other for the ratio of C3 to C1 (C3:C1) in lactate. Figure 4 contains lines of constant C1:C2 ratio, while Figure 5 contains lines of constant C3:C1.

Theoretically, ^{13}C NMR of the secreted lactate would permit the experimental determination of both C1:C2 and C3:C1. If C3:C1 is measured via NMR, the corresponding contour line from Figure 5 defines a relationship between f_1 and f_2. While the measurement of C3:C1 does restrict the flux ratios to a single contour line, determination of unqiue values for f_1 and f_2 requires the additional measurement of C2:C1. The intersection of the C1:C2 contour line (from Figure 4) and the C3:C1 contour line (from Figure 5) determines the single pair of values for f_1 and f_2 that is consistent with the observed isotope ratios (Figure 6). For example, suppose that C3:C1 is 1.94 and C1:C2 is 0.95. From Figure 5 one can see that a C3:C1 ratio of 1.94 means that f_2 can vary between 0 and 1, while f_1 can only take on values between approximately 2.5 and 5. The C1:C2 = 0.95 contour line (from Figure 4) intersects the C3:C1 = 1.94 contour line (from Figure 5) at a single point, with $f_1 = 0.25$ and $f_2 = 0.85$.

The modeling of isotope distribution indicates that a single experiment with ^{13}C-labeled glucose should be able to uniquely determine two critical flux ratios present in the energy metabolism of CRL-1606 hybridomas. The flux ratios determined from the *in vitro* ^{13}C NMR could be used by themselves to investigate hybridoma metabolism, or they could serve to validate the fluxes estimated independently through the application of material balances.

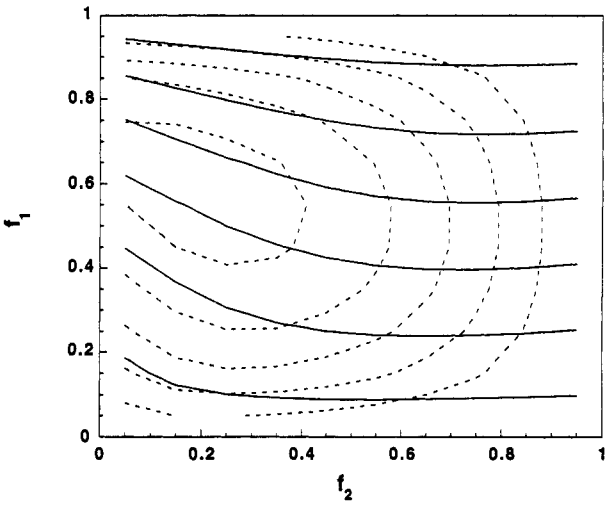

Figure 6. Superimposed contour plots to determine f_1 and f_2. The solid lines are contours of constant C3:C1 ratio (from Figure 4), and the dotted lines are contours of constant C1:C2 ratio (from Figure 3). Contour intersections determine f_1 and f_2.

Discussion

Model development for relating intracellular fluxes to isotope distribution is logically divided into two separate steps. The first is the generation of the stoichiometric equations that form the biochemical reaction network. The second step is to incorporate knowledge of specific reaction mechanisms to model the transfer of atoms from reactants to products. The method of modeling biochemical networks presented in this paper follows the same logical division. The vector representation of metabolite atom specific activities, combined with atom mapping matrices, enables the equations governing the isotopic steady state to be written directly from mass balance equations. The mass balances, in turn, follow directly from the set of stoichiometric reactions that forms the biochemical network. The derivation of the AMMs, which describe the transfer of atoms from reactants to products, is performed separately and is independent of the biochemical network. Thus, the development of the equa-

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tions relating intracellular fluxes and isotope distribution is simple and easy to understand. In addition, modifications to the network can be incorporated with virtually no algebra—facilitating rapid model development and refinement.

The method presented is readily applicable to the modeling and analysis of large and complex biochemical networks. The equations are easy to develop, modify, and understand, and their solution is rapid and requires only moderate computing power.

Appendix A: Gauss–Seidel Method

When a metabolic network is described using localized mapping matrices, the isotope distribution can be found iteratively. The specific activities of the substrates are initialized (to values between 0 and 1), and a consistent set of flux values must be provided. One by one each of the steady state equations is solved for its isotope distribution until convergence is achieved. This is essentially solving $\mathbf{As} = \mathbf{b}$ using the Gauss–Seidel method (*13*, *14*), in which $\mathbf{A}$ is split into two matrices $\mathbf{E}$ and $\mathbf{F}$ such that $\mathbf{A} = \mathbf{E} - \mathbf{F}$. Substitution and rearrangement yields

$$\mathbf{Es} = \mathbf{Fs} + \mathbf{b} \tag{A1}$$

The solution after the nth iteration, $\mathbf{s}_n$, is found in terms of the previous solution, s_{n-1}:

$$\mathbf{Es}_n = \mathbf{Fs}_{n-1} + \mathbf{b} \tag{A2}$$

The steady state equations, derived using atom mapping matrices, are essentially in the Gauss–Seidel form, although they are partitioned into separate matrix equations. Returning to the sample biochemical network of Figure 1, we will illustrate the relationship between the equations that result from using AMMs and the matrix equation obtained using the conventional method. Combination of eqs 21 and 22 into a single matrix equation gives

$$\underbrace{\begin{bmatrix} (V_2+V_4) & 0 & 0 & 0 \\ 0 & (V_2+V_4) & 0 & 0 \\ 0 & 0 & (V_3+V_6) & \\ 0 & 0 & 0 & (V_3+V_6) \end{bmatrix}}_{\mathbf{E}} \underbrace{\begin{bmatrix} \mathrm{AcCoA_I}(1) \\ \mathrm{AcCoA_I}(2) \\ \mathrm{AcCoA_{II}}(1) \\ \mathrm{AcCoA_{II}}(2) \end{bmatrix}}_{\mathbf{s}} = \underbrace{\begin{bmatrix} 0 & 0 & V_3 & 0 \\ 0 & 0 & 0 & V_3 \\ V_2 & 0 & 0 & 0 \\ 0 & V_2 & 0 & 0 \end{bmatrix}}_{\mathbf{F}}$$

$$\begin{bmatrix} \mathrm{AcCoA_I}(1) \\ \mathrm{AcCoA_I}(2) \\ \mathrm{AcCoA_{II}}(1) \\ \mathrm{AcCoA_{II}}(2) \end{bmatrix}_{\mathbf{s}} + \underbrace{\begin{bmatrix} V_1\mathrm{Pyr}(2) \\ V_1\mathrm{Pyr}(3) \\ V_5(\mathrm{Hex}(1)+\mathrm{Hex}(3)+\mathrm{Hex}(5)) \\ V_5(\mathrm{Hex}(2)+\mathrm{Hex}(4)+\mathrm{Hex}(6)) \end{bmatrix}}_{\mathbf{b}} \tag{A3}$$

Division of rows one and two by V_1 and rows three and four by V_5 and subtraction of $\mathbf{Fs}$ from both sides will yield eq 11. In practice, there is no need to combine the steady state equations into a single equation as nothing is gained. Instead, eqs 21 and 22 can be solved as written by the iterative method described above.

Notation

β_{ox}	enzyme catalyzing β-oxidation of hexanoate
A	matrix of coefficients from steady state equations
$\mathrm{AcCoA_I}$	acetyl-coenzyme A in pool I
$\mathrm{AcCoA_{II}}$	acetyl-coenzyme A in pool II
aKG	vector containing fractional enrichment of α-ketoglutarate carbons
AMM	atom mapping matrix
b	vector of specific activities of substrate atoms
f_1	fraction of the flux into pyruvate that comes from glycolysis
f_2	fraction of the flux into α-ketoglutarate that comes from glutamate
f6p	vector containing fractional enrichment of fructose 6-phosphate carbons
g6p	vector containing fractional enrichment of glucose 6-phosphate carbons
gap	vector containing fractional enrichment of glyceraldehyde 3-phosphate carbons
glc	vector containing fractional enrichment of glucose carbons
gln	vector containing fractional enrichment of glutamine carbons
Hex	hexanoate
lac	vector containing fractional enrichment of lactate carbons
mal	vector containing fractional enrichment of malate carbons
OAA	oxaloacetate
PD	pyruvate dehydrogenase
Pyr	pyruvate
r5p	vector containing fractional enrichment of ribose 5-phosphate carbons
s	vector of specific activities of the intermediate carbon atoms
s7p	vector containing fractional enrichment of sedoheptulose 7-phosphate carbons
V_i	flux through reaction i (in AcCoA network)
x_i	forward flux through reaction i (in hybridoma network)
x_{ir}	reverse flux through reaction i (in hybridoma network)
x_G	lumped flux of glutamate to α-ketoglutarate (in simplified hybridoma network)
x5p	vector containing fractional enrichment of xylulose 6-phosphate carbons
$[\mathbf{x}>\mathbf{y}]_n$	atom mapping matrix that describes the transfer of carbons from metabolite x to metabolite y in reaction n

Supplementary Material Available: Complete Mathematica program, including mapping matrix definitions, for modeling the simplified hybridoma metabolism (file name f1f2Net; 6 pages). Ordering information is given on any current masthead page.

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